Gursevak Singh Aulakh

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Professional Summary

Final-year Computer Engineering student with hands-on experience in **Artificial Intelligence, Android Development, Computer Vision, IoT, and Full-Stack Web Development**. Co-inventor of a **published Indian patent** on AI-powered missing person detection and family reunification for mass gatherings using hybrid identity fusion. Passionate about building scalable, real-world AI systems that integrate machine learning, embedded hardware, and cloud technologies to solve impactful societal problems.

Education

Guru Gobind Singh College of Engineering & Research Centre, Nashik — **2023 – 2026**
Bachelor of Engineering in Computer Engineering — *Savitribai Phule Pune University*

- **CGPA: 8.46/10**

Guru Gobind Singh Polytechnic, Nashik — **2020 – 2023**
Diploma in Computer Engineering — *Maharashtra State Board of Technical Education*

- **Percentage: 86.57%**

Guru Gobind Singh Public School, Nashik — **2019 – 2020**
Secondary School Certificate (SSC) — *Maharashtra State Board*

- **Percentage: 84.40%**

Projects

Kumbh Bandhu: Snan to Suraksha | ***Kotlin, Firebase, TensorFlow Lite, ArcFace CNN, ESP32, RFID, React***

- Architected an AI-powered missing person detection and family reunification platform for large-scale public gatherings, integrating Android, Edge AI, RFID, and cloud technologies; technology protected under a **published Indian patent (Application No. 202621047713 A).**
- Developed a real-time facial recognition pipeline using **ArcFace (512-dimensional embeddings)** with TensorFlow Lite, achieving **94.7% identification accuracy** under real-world conditions.
- Designed a hybrid identity fusion framework combining facial embeddings, RFID observations, gait signatures, and behavioral traits into a composite confidence score for reliable identification and reduced false positives.
- Implemented a complete Android-based volunteer coordination workflow with live RFID tracking, Firebase synchronization, QR verification, and location-aware emergency response, reducing manual search effort by approximately **73%**.
- Awarded **2nd Prize at National Level Techno Fest 2026** and shortlisted among the **Top 33 of 135+ teams** at Synergy 2026 conducted by The Institution of Engineers (India).

Fateh Excellence ERP & Web Platform | ***TanStack Start, React 19, Supabase, Tailwind CSS*** — Live Demo

- Architected and deployed a production-grade, local-first ERP and public web platform for a preschool & daycare, digitizing admissions, student management, fee collection, attendance, payroll, and institutional administration.
- Engineered a quarterly 4-installment fee system with oldest-first payment allocation, walk-in payment collection, printable A4 receipts and payslips, attendance-linked payroll, soft-delete archives, and audit trail logging.
- Built a protected Admin CMS for non-technical staff and a server-rendered, SEO-optimized public website with structured `schema.org/Preschool` data, achieving sub-350ms First Contentful Paint and deployment on Vercel.
- **Currently deployed and actively used** by Fateh Excellence Preschool & Daycare for daily institutional operations, administration, and public-facing engagement.

EVA: AI Institutional Desk Robot | ***Python, FastAPI, React, SQLite, ESP32-S3, Gemini AI, Kotlin***

- Built an AI-powered institutional desk robot integrating **Gemini AI, FastAPI, React, SQLite, and ESP32-S3** to provide conversational access to student, faculty, attendance, timetable, and departmental information.
- Developed an end-to-end cyber-physical system featuring tool-orchestrated AI queries, a web-based administration dashboard, bulk Excel data management, and real-time communication between backend services, the robot, and a Kotlin Android companion app.
- Engineered the physical prototype with voice interaction, dual OLED expressive eyes, I2S audio, embedded firmware, and WebSocket-based communication—combining AI, full-stack development, embedded systems, and robotics into a single institutional automation platform.

Granthalya – Sikh Granth Digital Library | ***React, TypeScript, Supabase, PostgreSQL, TanStack Query*** Live Demo

- Developed a full-stack digital library platform preserving Sikh literature with verse-wise reading, multilingual translations, and structured navigation across more than **218 chapters**.
- Built a Scholar's Workbench featuring a triple-column verse editor, editorial workflow (Draft → Review → Published), audio timestamp synchronization, and secure role-based access using Supabase Authentication.
- Designed a scalable hierarchical database architecture with PostgreSQL and optimized client-side caching using TanStack Query, enabling efficient retrieval of thousands of verses and metadata records.

Patent

Published Indian Patent – AI-Powered Missing Person Detection & Family Reunification System **Published Jun 2026**
Co-Inventor (1 of 7) | Indian Patent Application No. 202621047713 A *Filed Apr 2026*

- Co-invented a **published Indian patent** introducing a privacy-preserving, multi-modal identity recognition framework for locating missing persons and enabling family reunification in large-scale public gatherings.
- Designed a **hybrid identity fusion architecture** that combines facial embeddings, RFID observations, gait signatures, and behavioral traits into a unified confidence score, significantly improving identification reliability.
- The patented architecture serves as the technical foundation of the **Kumbh Bandhu** platform, integrating Edge AI, Android, IoT, and cloud technologies for real-time emergency response.

Experience

Data Science Intern **Jan 2025 – Mar 2025**
Acmegrade (Remote)

- Performed data preprocessing, feature engineering, and exploratory data analysis (EDA) using Python, Pandas, and NumPy to prepare datasets for machine learning workflows.
- Built and evaluated machine learning pipelines using Scikit-learn, applying data visualization and statistical analysis to derive actionable insights.
- Presented analytical findings through visual dashboards and reports, improving interpretation of model performance and business metrics.

Web Development Trainee **Jul 2022 – Aug 2022**
Calibers Infotech, Nashik

- Developed dynamic web applications using PHP, MySQL, HTML, CSS, and JavaScript following MVC-based development practices.
- Implemented user authentication, database connectivity, server-side form handling, and CRUD operations for responsive web applications.
- Collaborated with senior developers to debug applications, optimize SQL queries, and improve overall application performance.

Workshop Facilitator – AI Tools for Engineers **Jun 2026**
Department of Computer Engineering, Guru Gobind Singh Polytechnic, Nashik

- Conducted a two-day hands-on workshop for students and faculty on Generative AI, Prompt Engineering, AI-assisted software development, and productivity tools.
- Delivered live demonstrations using ChatGPT, Claude, Gemini, Perplexity, NotebookLM, Gamma, and Lovable for coding, research, documentation, presentation creation, and rapid application development.
- Mentored participants in building AI-powered workflows and practical projects, enabling effective adoption of modern AI tools in academic and engineering applications.

Technical Skills

Programming Languages: Java, Kotlin, Python, C++, JavaScript, TypeScript, PHP, SQL

Mobile Development: Android SDK, Jetpack Components, Firebase Authentication, Firebase Realtime Database, Firebase Storage, Cloud Functions

Web Development: React, TanStack Start, TanStack Router, TanStack Query, Vite, HTML5, CSS3, JavaScript (ES6+), TypeScript, PHP, REST APIs, Tailwind CSS, Radix UI

Artificial Intelligence & Machine Learning: TensorFlow Lite, ArcFace, Computer Vision, OpenCV, Tesseract OCR, Scikit-learn, Pandas, NumPy, Feature Engineering, Exploratory Data Analysis (EDA)

Databases & Cloud: Firebase, PostgreSQL, MySQL, Supabase, Vercel

Embedded Systems & IoT: ESP32, RFID (HF/UHF), Edge AI, Real-Time Device Communication

Developer Tools: Git, GitHub, Android Studio, VS Code, Jupyter Notebook

Core Computer Science: Object-Oriented Programming, DBMS, Operating Systems, Computer Networks, RESTful APIs

Achievements & Leadership

National Level Techno Fest 2026 – Second Prize **Apr 2026**

L.G.N. Sapkal College of Engineering, Nashik

- Awarded **2nd Prize** for developing the AI-powered **Kumbh Bandhu** platform, competing against engineering teams from multiple institutions.

Synergy 2026 – Top 33 Finalist **Apr 2026**

The Institution of Engineers (India), Nashik Local Centre

- Selected among the **Top 33 teams out of 135+** entries for presenting an innovative AI-enabled crowd safety solution for mass gatherings.

AI Workshop Facilitator **Jun 2026**

Guru Gobind Singh Polytechnic, Nashik

- Conducted a two-day hands-on workshop on Generative AI, Prompt Engineering, and AI-assisted software development for students and faculty.

Research Paper Presentation **Jun 2023**

ICRTETM 2023

- Presented the paper titled *"Animal Rescue and Wellness Web Application"* at the Second International Conference on Recent Trends in Engineering, Science, Technology and Management.

Telecom Technician – IoT Devices/Systems (Level 4) **Sept 2023**

Telecom Sector Skill Council (TSSC) *Grade: B*

- Successfully completed the NSQF-certified IoT Devices/Systems training program recognized by NCVET.

Additional Information

Languages: English (Professional), Hindi (Native), Punjabi (Native)

Areas of Interest: Artificial Intelligence, Computer Vision, Android Development, IoT, Embedded Systems, Full-Stack Development

Soft Skills: Leadership, Public Speaking, Technical Mentoring, Problem Solving, Team Collaboration